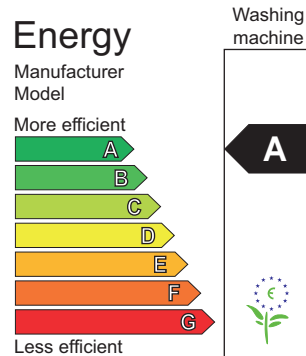


Answer all questions.

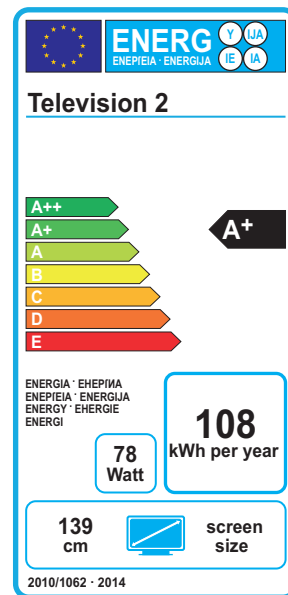
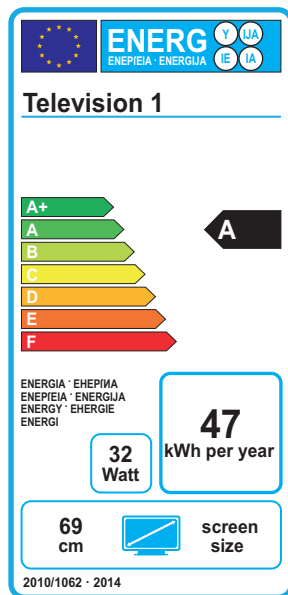
1. Energy rating labels are compulsory on most household appliances such as fridges, dishwashers, washing machines and televisions. These labels allow customers to compare appliances. In addition, the labels give other information about the appliances such as how noisy they are.

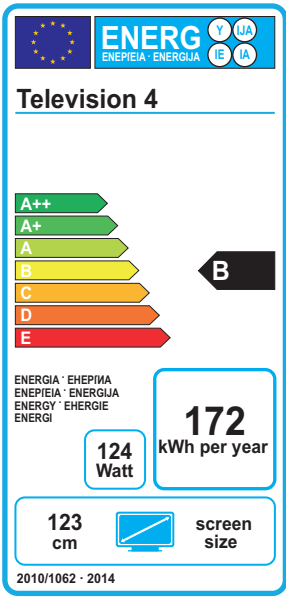
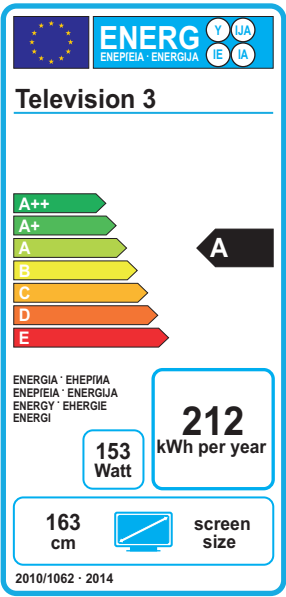
Labels used to rate appliances are from A, the most efficient, to G, the least efficient.



As manufacturers have designed more efficient devices, new categories, e.g. A+, A++ and A+++ have been added to the labels.

The energy rating labels of four televisions are given below and opposite.





Information on the cost of each television is given below.

Television	Purchase cost (£)
1	280
2	1 000
3	1 500
4	800

(a) Use the information from the labels and the table to tick (✓) the **three** correct statements below. [3]

- Television 1 uses less energy per second than television 2
- The largest televisions always use the most energy
- The purchase cost of television 2 is 1.5 times that of television 3
- More expensive televisions always use less energy
- Television 3 uses 40 units more per year than television 4
- Televisions with the same energy rating, e.g A+, don't always have the same power



- (b) It is claimed that power is proportional to screen size. Use the data for television **1** and television **2** to determine if this claim is true. [3]
Space for calculations.

.....

- (c) (i) Use the equation:

$$\text{time (h)} = \frac{\text{units used (kWh)}}{\text{power (kW)}}$$

and data from the energy label to calculate how many hours the label suggests that television **2** is used for in 1 year. [2]

Hours used =

- (ii) Use an equation from page 2 to calculate the cost of using television **2** for 1 year if 1 unit (kWh) of electricity costs 16 p. Give your answer in pounds (£). [2]

Cost = £



- (iii) The expected lifespan of a television is 10 years. Simon concludes that it will be more cost effective to buy and run television **2** for 10 years but Sarah disagrees and claims that television **4** will be cheaper. Use the data to determine who is right. [3]
Space for calculations.

- (iv) Other than to save money, why should consumers be encouraged to choose appliances that use less energy? [2]

